



Constellation

Clinton Power Station
8401 Power Road
Clinton, IL 61727

U-604815
September 24, 2024

10 CFR 50.73

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Clinton Power Station, Unit 1
Facility Operating License No. NPF-62
NRC Docket No. 50-461

Subject: Licensee Event Report 2023-002-01

Enclosed is Licensee Event Report (LER) 2023-002-01: Reactor Core Isolation Cooling Inoperable Prior to Mode Change, Revision 1. This report is being submitted in accordance with the requirements of 10 CFR 50.73.

There are no regulatory commitments contained in this report.

Should you have any questions concerning this report, please contact Mr. David Livingston, Regulatory Assurance Manager, at (217) 937-2800.

Respectfully,

Norha Z. Plumey
Site Vice President
Clinton Power Station

IEZZ
NRR

Attachment: Licensee Event Report 2023-002-01

cc:

Regional Administrator - Region III
NRC Senior Resident Inspector - Clinton Power Station
Office of Nuclear Facility Safety - Illinois Emergency Management Agency



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Clinton Power Station



050



052

2. Docket Number

00461

3. Page

1 OF 3

4. Title

Reactor Core Isolation Cooling System Inoperable Prior to Mode Change

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
12	19	2023	2023	- 002 -	01	09	24	2024	Facility Name	☐ 052	Docket Number

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

David Livingston, Regulatory Assurance Manager

Phone Number (Include area code)

(217) 937-2800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected



No



Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month

Day

Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On October 10, 2023, at 23:02 CST, the Reactor Core Isolation Cooling (RCIC) system tripped during performance of Surveillance Requirement (SR) 3.5.3.3. This SR is performed when RCIC steam supply pressure is at the higher operating ranges of the system to verify that the RCIC pump can develop the required flow rate against a system head corresponding to reactor pressure. The system was restored to operable on October 16, 2023, at 02:51 CST following replacement and calibration of the Electric Governor-Remote (EG-R). Engineering personnel suspected that the RCIC system trip was due to a faulty EG-R; however, subsequent failure analysis identified no significant issues with the EG-R. On December 19, 2023, Engineering personnel determined that work performed on August 28, 2023 affected functionality of the EG-R and caused RCIC to be inoperable per Technical Specifications (TS) 3.5.3, "RCIC System."

This event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS 3.0.4. Specifically, on October 9, 2023, at 05:14 CST, during unit startup following a refueling outage, the Applicability for TS 3.5.3 was entered (i.e., Mode 2 with reactor steam dome pressure greater than 150 psig) with the RCIC system inoperable. In addition, the unit was taken to Mode 1 on October 10, 2023.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Clinton Power Station	☒ 050	2. DOCKET NUMBER 00461	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER - 002	REV NO. - 01

NARRATIVE**CONDITION PRIOR TO THE EVENT**

Unit: 1
Mode: 1

Event Date: December 19, 2023
Mode Name: Power Operation

Event Time: 0800 CST
Reactor Power: 100

EVENT DESCRIPTION

On October 8, 2023, Operations commenced reactor startup following a refueling outage. On October 10, 2023, at 23:02 CST, the Reactor Core Isolation Cooling (RCIC) system tripped on low suction pressure during performance of Surveillance Requirement (SR) 3.5.3.3 and the RCIC system was declared inoperable. This SR is performed when RCIC steam supply pressure is at the higher operating ranges of the system to verify that the RCIC pump can develop the required flow rate against a system head corresponding to reactor pressure. Troubleshooting activities commenced to determine the cause of the RCIC system trip. As part of the troubleshooting activities, the RCIC Electric Governor – Remote (EG-R) was replaced and calibrated. Following replacement of the EG-R, SR 3.5.3.3 was re-performed satisfactorily and the RCIC system was declared operable at 02:51 CST on October 16, 2023.

Based on the successful re-performance of SR 3.5.3.3 following replacement of the EG-R, Engineering personnel suspected that the RCIC system trip on low suction pressure was due to a faulty EG-R. The suspect EG-R was sent offsite for failure analysis. Results from the failure analysis were documented in a failure analysis report that was issued on November 17, 2023. The report stated that the EG-R showed no signs of degradation and that the sharp, rapid changes in the EG-R output that were observed on October 10, 2023, could not be replicated.

Efforts continued to determine the cause of the RCIC system low suction pressure trip. On December 19, 2023, a review of Plant Process Computer data revealed an anomaly to the Electric Governor – Magnetic Pickup (EG-M) output that occurred on August 28, 2023. Further investigation determined that on August 28, 2023, with Unit 1 in Mode 1 at 100% power, work associated with a plant modification on the transient test system disturbed electrical connections for the EG-M. Transient Test is a high-speed data acquisition system which monitors and records analog and digital signals from various plant systems. The EG-M is the flow control system component that transmits an electrical signal to the EG-R proportional to the error signal developed between turbine speed demand and actual turbine speed. When the electrical connections were disturbed, EG-M output fell to 0% scale, indicating a loss of functionality of that component. This was unrecognized by station personnel at the time of occurrence.

Further investigation has also found that between September 11, 2023 and October 8, 2023, through various maintenance activities performed on the RCIC system, there is a potential for air to have been entrained in the lubricating oil system. Air entrainment in the lubricating oil control subsystem can cause governor speed and ramp control issues.

On December 19, 2023, at approximately 08:00 CST, it was determined that the work performed on August 28, 2023, affected functionality of the EG-M and potentially caused the RCIC system to be inoperable per Technical Specifications Section 3.5.3, "RCIC System." Condition A of TS 3.5.3 applies when the RCIC system is inoperable, with a Required Action to restore the RCIC system to operable status within 14 days. The Completion Time for this Required Action was not exceeded because the Applicability for TS 3.5.3 was exited prior to the expiration of the 14-day Completion Time to support the unit shutdown for a planned refueling outage.

However, this event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS 3.0.4. Specifically, on October 9, 2023, at 05:14 CST, during unit startup following the refueling outage, the Applicability for TS 3.5.3 was entered (i.e., Mode 2 with reactor steam dome pressure greater than 150 psig) with TS 3.5.3 not met because the RCIC system was inoperable. In addition, the unit was taken to Mode 1 on October 10, 2023.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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YEAR	SEQUENTIAL NUMBER	REV NO.					
2023	- 002	- 01					

NARRATIVE**CAUSE OF THE EVENT**

The probable cause of the RCIC Turbine Trip was either:

1. An electrical disturbance on the EG-M resulting from the Transient Test system modification that resulted in a loss of control output to the EG-R
2. Outage activities (system cooldown and tagout blocking) that resulted in air formation in the RCIC lubricating oil system

SAFETY CONSEQUENCES

The condition described in this LER is reportable under 10 CFR 50.73(a)(2)(i)(B), any operation or condition which was prohibited by the plant's TS, due to entering the Applicability for TS 3.5.3 with the RCIC system inoperable. There were no actual safety consequences associated with the event. The RCIC system is not an engineered safety feature and no credit is taken in the safety analysis for its operation. Furthermore, RCIC does not meet the threshold for a failure of a single train safety system since it is not included in the safety analysis. High Pressure Core Spray (HPCS) system remained operable and therefore able to perform its safety function during the entire period that RCIC was inoperable. There were no actual demands for either RCIC or HPCS during this period.

CORRECTIVE ACTIONS

The EG-R was replaced and the replacement unit calibrated in accordance with the Electric Power Research Institute (EPRI) Terry Turbine Maintenance Guide.

Update maintenance procedures to expand direction to fill and vent lube oil subsystem following all forms of maintenance

Update maintenance procedures to expand direction to calibrate the EG-M following any work that could result in even minor changes to the RCIC governor electrical circuit

Implement BWR Owners Group recommendation for a modification to insert a 7 second time delay on RCIC low suction pressure.

PREVIOUS OCCURENCES

A review of Licensee Event Reports for the past five years identified no previous similar occurrences at Clinton Power Station.